AVIAN PAIN MANGEMENT AND ANESTHESIA

David Sanchez-Migallon Guzman,
LV, MS, Dipl. ECZM (Avian, Small Mammal), Dipl. ACZM
Companion Exotic Animal Medicine and Surgery,
School of Veterinary Medicine University of California Davis
One Shields Avenue 2108 Tupper Hall Davis, California 95616
Email: guzman@ucdavis.edu

1. PAIN MANAGEMENT

1.1. PAIN RECOGNITION AND ASSESSMENT

Pain is defined as the sensory and emotional experience associated with actual or potential tissue damage. Pain is subjective and has an emotional component that can be difficult for us to translate because most avian species lack facial expression and do not share verbal language with humans. Nociception is the transduction, conduction, and central nervous system processing of signals generated by the stimulation of nociceptors. It is the physiologic process that, when performed to completion in a conscious animal, results in the perception of pain.

Behavioral changes associated with pain in birds can be subtle. Behavioral changes do manifest differently among different species of birds, and observers must become familiar with the full range of normal behaviors for the species and the individual. Birds tend to respond to noxious stimuli with a fight-or-flight response (i.e., escape reactions, vocalization, excessive movement) and/or conservation–withdrawal responses (i.e., no escape attempts or minimal vocalization and immobility). In domestic chickens, removal of feathers caused a progression of behavioral changes from an initial alert-agitated response to periods of crouching immobility following successive removal of feathers.[1] Movement, head motions, and beak clacks were all significantly reduced in red-tailed hawks hospitalized with recent orthopedic injuries, compared with birds without recent orthopedic injuries.[2] Cockatiels injected with carrageenan in the plantar surface exhibited significantly increased focal preening and increased burst preening, while control cockatiels spent significantly more time in an upright stance.[3] Social interactions might decrease in birds that have social systems. Birds in pain may display guarding behavior to protect a painful area. Grooming activity may decrease when a bird is in pain; conversely, over grooming and feather-damaging behavior have been associated with chronic pain, which may include neuropathic pain.

Pain scales and score sheets are tools increasingly used to assess pain in animals, especially when specifically designed for a given species under well-defined conditions. Using pain scales requires understanding normal and pain-related behavior for the species and individual. Pain score sheets can help improve uniformity and efficacy of pain scoring using behavioral analysis. Score-sheet descriptions of behavior must be refined, and terms must be clearly defined to reduce observer bias and interobserver variability. A series of behavioral and postural parameters that could be considered includes mentation, environmental interaction, perching, maintenance behaviors, locomotion, focal preening of singular area, body posture and appetite.[4] In lieu of species-specific pain score sheets, a generic pain scale of 1 to 10 is useful for assessing bird's pain and the response to treatment and recovery from a painful condition. In a study using pigeons with experimental fractures, two numeric pain scales (scores from 0 to 5) were sensitive and specific for detecting pain. The first evaluated the position of the fractured limb (from 0, bearing the same weight on both feet, to 5, lying on the floor) and the second evaluated the overall assessment of the bird (from 0, no sign of pain, to 5, unable to stand, lying on the floor).[5] Effective analgesia is expected to show a marked, easily discernable change in posture or behaviors that will effect a reliable change in the subjective pain score. If no change in pain score occurs, then the drugs, dosage, or frequency of administration should be re-evaluated for that patient.

1.2. TREATMENT OF PAIN

Pre-emptive analgesia with administration of analgesics prior to tissue injury will prevent central and peripheral sensitization. Pre-emptive analgesia with opioid drugs, nonsteroidal anti-inflammatory drugs (NSAIDs), and/or local anesthetics can block transmission of sensory noxious stimuli to the central nervous system, which will reduce the overall potential for inflammation and pain and improve the patient's short-term and long-term recovery. Multimodal protocols with opioids, NSAIDS, local anesthetics, and/or other drugs acting at different points in the nociceptive system provide a greater effect and potentially less toxicity than individual drugs given alone.

1.2.1. Opioid Drugs

Butorphanol tartrate, a κ-opioid receptor agonist and μ-opioid receptor antagonist, is currently considered the analgesic drug of choice for management of acute pain in psittacines. Previous studies have evaluated the analgesic effects of opioids, particularly those with κ opioid receptor affinities, as well as their effects on anesthetic requirements in psittacine birds. The accepted dose for butorphanol tartrate in psittacines of 1 to 5 mg of butorphanol/kg IM requires repeated administration every 2 to 3 hours.[6,7] In a pharmacokinetic study in Hispaniolan Amazon parrots (*Amazona ventralis*),[8] mean target plasma concentrations greater of 100 ng/mL were maintained for 1.5–2 hours and 2–3 hours after intravenous and intramuscular administration, respectively. The termination half-life for was 0.49 hours and 0.51 hours for intravenous and intramuscular administration, respectively, and was well distributed with a rapid clearance. The calculated bioavailability of butorphanol administered intravenously and orally was 130% and 5.9%, respectively. The low oral bioavailability would limit the use of this route of administration for clinical purposes. Based on these results, in Hispaniolan Amazon parrots, butorphanol tartrate dosed at 5 mg/kg intravenous or intramuscular would have to be administered every 2 and 3 hours, respectively, to achieve plasma concentrations consistent with published therapeutic levels. Long-acting opioid drugs like liposomal-encapsulated butorphanol provided analgesia for up to 5 days in Amazon parrots.[9] Unfortunately, those formulations currently are not commercially available. Pre-operative butorphanol administration (2 mg/kg IM) did not show significant anesthetic or cardiopulmonary changes in Hispaniolan Amazon parrots anesthetized with sevoflurane, suggesting that it may be useful as part of a pre-emptive analgesic protocol.[10] In American kestrels,[11] butorphanol tartrate at 1, 3 and 6 mg/kg did not significantly increase foot withdrawal thermal thresholds compared to saline administration or baseline values. Neither did it result in sedative effects in American kestrels, but instead caused hyperaesthesia or hyperalgesia and agitation in males receiving 6 mg/kg. The study concluded that butorphanol may not provide clinically effective analgesia in American kestrels, but recommended further studies with other type of stimulation, formulations, dosages, routes of administration and testing times are needed to fully evaluate the antinociceptive and adverse effects of butorphanol in American kestrels and other species of birds and their relevance for clinical application. In chickens,[12] in a recent study butorphanol at 2 mg/kg IM have shown to provide analgesia for up to two hours.

Hydromorphone hydrochloride, a pure μ-opioid agonist drug, has shown a dose responsive thermal antinociception effect when administered IM at 0.1-0.6 mg/kg in American kestrels (*Falco sparverius*), suggesting that would produce analgesia in this species for up to 6 hours.[13] No significant differences in mean sedation-agitation scores were detected between hydromorphone and saline solution treatments; however, appreciable sedation was detected in 4 birds when administered at 0.6 mg of hydromorphone/kg. When the sedated birds were handled, they appeared agitated for several seconds but then resumed a sedated behavior. In a similar study in cockatiels (*Nymphicus hollandicus*) hydromorphone at 0.1, 0.3, and 0.6 m/kg did not significantly change the thermal foot withdrawal response despite plasma drug concentrations considered therapeutic for other species, but caused mild sedative effects at 0.3 and 0.6 mg/kg,[14] but higher doses of 1 and 2 mg/kg had a small but significant effect without mild or no sedation.[15] More recently, in a study in orange-winged Amazon parrots, hydromorphone at 1 and 2 mg/kg significantly increased thermal foot withdrawal response between 0.5 and 3 hours and 1.5 and 6 hours respectively, compared with the response after administration of saline. The lowest dose evaluated, 0.1 mg/kg did not have a significant effect. Significant agitation was observed for 1 and 2 mg/kg hydromorphone doses at 0.5, 1.5 and 3h. Other significant adverse effects when compared with control included nausea-like behavior, ataxia and pupillary constriction One bird vomited after receiving the 2 and 1 mg/kg dose at 0.5 and 1.5 hours respectively. While hydromorphone at 1 and 2 mg/kg IM appears to have desired analgesic effects, the presence of agitation, nausea-like behavior and ataxia warrants consideration of lower doses for clinical use in orange-winged Amazon parrots.

Fentanyl is a pure μ-opioid agonist. Fentanyl 0.02 mg/kg IM did not affect the withdrawal thresholds to electrical or thermal stimuli of white cockatoos (*Cacatua alba*), however, a tenfold increase in the dosage (0.2 mg/kg SC) did produce an analgesic response but many birds were hyperactive for the first 15 to 30 min after receiving the high dose.[16] In a recent study in red-tailed hawks (*Buteo jamaicensis*),[17] mean ± SD fentanyl plasma concentrations and isoflurane minimum anesthetic concentrations (MACs) were 0 ± 0, 8.51 ± 4, 14.85 ± 4.82 and 29.25 ± 11.52 ng/mL, and 2.05 ± 0.45%, 1.42 ± 0.53%, 1.14 ± 0.31% and 0.93 ± 0.32% for the target concentrations of 0, 8,16 and 32 ng/mL, respectively. At these concentrations fentanyl significantly decreased isoflurane MAC by 31%, 44% and 55%, respectively. Interestingly, the dose had no significant effect on heart rate, systolic, diastolic or mean arterial blood pressure. The study concluded that fentanyl produced a dose-related decrease of isoflurane MAC with minimal effects on measured cardiovascular parameters in red-tailed hawks. More recently, in a study with Hispaniolan Amazon parrots *(Amazona ventralis)*, fentanyl administered as an intravenous CRI reduced isoflurane MAC; however, a relatively very high dose was needed and a was accompanied by a depressive effect on heart rate and blood pressure that would need to be considered for application of this technique in clinical settings.[18] In a study including chickens of Brahma, Delaware, Redstar and Wyandotte breeds, the plasma concentrations of fentanyl were evaluated following application of 25 µg/hour transdermal fentanyl patches (TFP).[19] The TFPs were in place for 72 hours over the left iliopsoas muscle after plucking the feathers. There was a significant degree of inter-individual difference in maximum plasma fentanyl concentrations that was attributed to absorption variability, but all chickens reached the human target plasma concentrations of 0.2-1.2 ng/mL within 2-4 hours and had rapid decrease following removal of the patch. Based on data from this pharmacokinetic study, fentanyl transdermal application of patches for chickens targeted to provide 8–10 μg/kg per hour of fentanyl resulted in fentanyl plasma concentration within the range considered therapeutic in people for 72 hours.[19] However, because of substantial individual variation in absorption, results from longer applications would likely be inconsistent among patients.

Buprenorphine hydrochloride is μ-opioid receptor agonist but its κ-receptor activities are less well defined. Several studies suggest that buprenorphine demonstrates κ-receptor agonist activity but other evidence in mammals and pigeons (*Columba livia*) suggests that it also displays some κ antagonistic activities. Buprenorphine has unusual receptor-binding characteristics that seem to be the result of slow drug dissociation from opioid receptors. Few studies have been published evaluating the use of buprenorphine in birds. Buprenorphine at 0.1 mg/kg IM in grey parrots (*Psittacus erithacus*) did not increase withdrawal thresholds and latencies when an electrical noxious stimulus was applied,[20] but in a later study pharmacokinetic analysis showed that this dose did not achieve plasma concentrations considered to be effective in humans.[21] In American kestrels,[22] buprenorphine hydrochloride caused a significant thermal antinociceptive response at 0.1, 0.3 and 0.6 mg/kg for up to and over 6 hours compared to control treatment. At 0.6 mg/kg, a mild sedative effect was appreciated. The increase in mean withdrawal threshold was suggestive of analgesia. In a similar study in cockatiels,[23] buprenorphine at 0.6, 1.2, and 1.8 mg/kg did not significantly change the thermal foot withdrawal response, compared with the response for the control treatment. No significant change in agitation-sedation scores was detected between all doses of buprenorphine and the control treatment. A concentrated and also a sustained-release buprenorphine concentration have also been evaluated recently in American kestrels[24,25] and red-tailed hawks,[26] requiring less frequent administration and maintaining drug plasma concentrations for a longer period of time. In orange-winged Amazon parrots, the thermal antinociceptive effects of a concentrated formulation of buprenorphine (Simbadol®) after SC administration was evaluated.[27] Buprenorphine at 2 mg/kg resulted in small but significant increase in thermal threshold when compare to saline, while lower doses of 0.1 and 1 mg/kg did not result in a significant effect. No significant effect of buprenorphine on agitation-sedation score was observed at any of the doses evaluated. These results provide evidence of weak efficacy at lower doses for pain management in this species, similar to previous findings in the cockatiels, and highlights the differences in efficacy with the raptor species evaluated. Further studies are needed to fully determine the analgesic and adverse effects of buprenorphine in psittacines and warrants evaluation in other species.

Tramadol hydrochloride, a centrally acting μ-opioid receptor agonist drug that binds weakly to κ- and □- opioid receptors, also inhibits the reuptake of norepinephrine and serotonin. Tramadol at a dose of 30 mg/kg orally, induced thermal antinociception in Hispaniolan Amazon parrots. This dose was necessary for induction of significant and sustained effects with duration of action up to 6 hours, while lower dosages of 10 and 20 mg/kg did not have a significant effect.[28] The pharmacokinetics of tramadol have been evaluated in several avian species, including bald eagles (*Haliaeetus leucocephalus*), red-tailed hawks, peafowl (*Pavo cristatus*), African penguins (*Spheniscus demersus*), Muscovy ducks (*Cairina moschata domestica*) and also in Hispaniolan Amazon parrots.[29] The results of these studies detected differences in pharmacokinetic between Hispaniolan Amazon parrots and other species of birds. Low oral bioavailability, only 23.48% in parrots compared with 97.94% in eagles,[30] and rapid clearance of tramadol led to a requirement for a higher dose for oral administration than that reported for other species. More recently, the thermal antinociceptive properties of tramadol hydrochloride have also been evaluated in American kestrels.[31] The results of the study have shown that oral tramadol at 5 mg/kg significantly increased thermal nociception thresholds for 1.5 hours when compared to control values and for 9 hours when compared to baseline thresholds, while 15 and 30 mg/kg resulted in less antinociceptive effects. In Muscovy ducks (*Cairina moschata domestica*), tramadol at 30 mg/kg PO was effective in improving a number of variables in a temporary arthritis model induced by an injection of 3% monosodium urate solution into the intertarsal joint and assessed by ground-reactive forces measured by a pressure-sensitive walkway (PSW) system.[32]

1.2.2.Nonsteroidal Anti-inflammatory Drugs

Meloxicam is a COX-2 selective oxicam NSAID. Meloxicam is currently available as oral tablets that are made into suspension, and an injectable form. In Hispaniolan Amazon parrots with experimental arthritis,[33] a dosage of 1 mg/kg meloxicam every 12 hours IM was necessary to achieve significant return to baseline weight bearing. In pigeons,[34] administration of meloxicam at 0.5 mg/kg every 12 hours orally appeared ineffective in minimizing postoperative orthopedic pain with an experimental femoral osteotomy, but the 2.0 mg/kg dose provided quantifiable analgesia that appeared safe in this species. The pharmacokinetic profile has been evaluated in a large number of species that includes raptors, psittacines, Columbiformes, Anseriformes, Galliformes, and ratites. [35,36,37,38,39,40] In most of the species in which PK has been determined, the oral bioavailability ranges from approximately 50%-75%, but other pharmacokinetic differences between species supports the need for species-dependent studies and underlines the challenges of extrapolating drug dosages between species. Two clear exceptions are the Grey parrots[40] and the cockatiels,[41] where the oral bioavailability has been determined to be 38 and 11%. In addition, the adverse effects of meloxicam have been evaluated in species of psittacines,[42,43,44] Galliformes[45] and kestrels.[46] In these studies, there were minimal to no renal, gastrointestinal or hemostatic adverse effects reported at the therapeutic doses evaluated. As an example, the safety of oral meloxicam in Hispaniolan Amazon parrots following administration of 1.6 mg/kg for 14 days was evaluated. Parameters for renal, gastrointestinal, and hemostatic evaluation were obtained from whole blood, serum, fecal, and urine samples prior to treatment and at select time points following initiation of treatment. N-acetyl-b-D-glucosaminidase (NAG) was used as a marker for renal toxicity in both urine and serum. Urine was screened for casts, specific gravity, protein, glucose, ketones, pH, and hemoglobin; feces were screened for occult blood. Serum was screened for NAG, glutamate dehydrogenase, alanine transaminase, aspartate aminotransferase, total protein, albumin, and cholesterol. Whole clotting time was also assessed. There was no evidence of significant renal, gastrointestinal, or hemostatic effects at the dosage evaluated. In grey parrots[47] evaluating multiple dose administration, a 1mg/kg PO q 24 h dose resulted in significant plasma drug accumulation and significant elevation of NAG, suggesting that lower doses might be required and acknowledging the need for further studies. Adverse effects have also been evaluated in Asian vultures (*Gyps bengalensis*, *G. indicus*, and *G. tenuirostris*), in which renal toxicity has been a reported with other NSAIDS [48,49] but not with meloxicam.[50] A survey to determine NSAID toxicity in captive birds treated in zoos reported no fatalities when meloxicam was administered to over 700 birds from 60 species [51]. Despite these reports, the general recommendations regarding safety of NSAIDS in birds still need to be considered when using meloxicam. Because of the efficacy, PK, and safety data available, meloxicam is the NSAID of choice to treat perioperative and chronic pain. A sustained-release formulation of meloxicam (Meloxicam SR, Wildlife Pharmaceuticals, Windsor, CO, USA) has been evaluated in Hispaniolan Amazon parrots[52] and American flamingos.[53] In Hispaniolan Amazon parrots, this formulation results in a rapid rise to peak concentrations and maintains above target plasma concentrations for 12 to 96 hours after a single dose of 3 mg/kg SC, likely due to interindividual variability. In American flamingos, the sustained released meloxicam formulation had also a rapid rise to peak plasma concentration, but only remained above target plasma concentrations for 0.5 to 4 hours after a single subcutaneous dose of 3 mg/kg.

Grapiprant is a prostaglandin E2 (PGE_2) receptor antagonist NSAID approved for use in dogs in the US. Grapiprant binds to the EP4 receptor, blocking the sensitization of sensory neurons and stimulation of inflammation, mediated by the PGE_2 receptor. It does not rely on inhibition of COX enzymes and preserves functions associated with COX enzymes pathways. Oral administration of a single dose of grapiprant at 30 mg/kg to fasted red-tailed hawks reached plasma concentrations 19-32 times the canine therapeutic concentrations within 2 hours approximately and with no adverse effects identified.[54] Grapiprant administered at the same dose to the same RTHAs with food had lower plasma concentrations by 88%, with delayed absorption and a similar half-life.[55] Grapiprant plasma concentrations were achieved above the canine therapeutic concentrations within 16 hours postmedication. Mean concentrations were maintained for approximately 20 hours. Simulations support a dosing frequency of 12-hour intervals with food reaching minimum effective concentrations established for canines, although it is unknown whether these plasma concentrations are therapeutic for birds.[55] Further studies to evaluate grapiprant pharmacokinetics, efficacy and safety in red-tailed hawks and other avian species are needed.

1.2.3. Local Anesthetics

Local anesthetics reversibly bind to Na^+ channels and block impulse conduction. Local anesthetics will be absorbed by the vasculature in the region being blocked. Systemic uptake of the local anesthetics can be rapid in birds, and metabolism may be prolonged, increasing the potential for toxic reactions. Dosage recommendations have been lower for birds than mammals because anecdotal evidence has suggested that birds may be more sensitive to the adverse effects of these drugs. Toxic effects reported in birds include fine tremors, ataxia, recumbency, seizures, stupor, cardiovascular effects, and death. No adverse cardiovascular effects were identified in a study in chickens under isoflurane anesthesia when lidocaine was administered at 6 mg/kg IV.[56]

Lidocaine is available as a commercial preparation of 2% (20 mg/mL) and has a relatively short duration of action. Based on empirical use, the recommended dosage is 2 to 4 mg/kg. For small birds, the commercial preparation may need to be diluted 1:10 or more to achieve an effective volume for the block, but it is unknown whether dilution allows either appropriate tissue drug concentrations for analgesia or the expected duration of analgesia to occur. The PK profile of lidocaine following 2.5 mg/kg IV administration in anesthetized chickens showed a short half-life and appears to share similar mechanisms of metabolism and elimination to mammalian species reported.[57] No adverse cardiovascular effects were identified in a study in chickens under isoflurane anesthesia when lidocaine was administered at 6 mg/kg IV.[56] Neither a low dose (3 mg kg^{-1} followed by 3 mg kg^{-1} $hour^{-1}$) or a high dose (6 mg kg^{-1} followed by 6 mg kg^{-1} $hour^{-1}$) of lidocaine was administered IV had an effect in the MAC in chickens.[58]

Bupivacaine is available commercially as 0.25, 0.5, and 0.75% solutions (2.5, 5, and 7.5 mg/mL, respectively), and the lower concentration may not need dilution for birds. It has been used conservatively in birds due to concerns for toxic effects. Based on empirical use, the recommended maximum dosage of bupivacaine is 2 mg/kg by the author. It has shown efficacy in chickens following intra-articular administration (3 mg in 0.3 mL saline) to treat arthritic pain[59] and topically for pain associated with amputated beaks (1:1 mixture of bupivacaine and dimethyl sulfoxide) to improve feed intake.[60] The PK profile has been determined in ducks at 2 mg/kg SC suggesting a shorter duration of action than in mammals but with potential for delayed toxicity.[61] The cardiovascular tolerance of bupivacaine was evaluated in chickens anesthetized with isoflurane.[62] In this study it was determined that 1.96 mg/kg IV bupivacaine is associated with a 50% probability of a clinically relevant change in MAP and HR, whereas 1.33 mg/kg IV bupivacaine is associated with a 1% probability of causing the same effect.

Local line or splash blocks are the most common methods of regional infiltration used in birds. The subcutaneous space in most avian species is very thin so a small gauge needle is recommended to make several SC injections into the operative area. Brachial plexus blockade has been described in a variety of avian species but neither bupivacaine (2 and 8 mg/kg) or lidocaine (15 mg/kg) with epinephrine perineurally effectively blocked nerve transmission in the brachial plexus in ducks [63] or in chickens with lidocaine (20 mg/kg) or bupivacaine (5 mg/kg) with epinephrine for brachial plexus blockade using a nerve locator.[64] In Amazon parrots, neither palpation nor ultrasound-guided brachial plexus blockade was found to result in an effective block using lidocaine at 2 mg/kg perineurally.[65] Sciatic-femoral nerve block has also been described in falcons, and was considered a feasible technique by applying lidocaine at 2 mg/kg perineurally with the aid of a nerve locator.[66] In broiler chickens, spinal block has been recently evaluated with administration of lidocaine and bupivacaine into the synsacrococcygeal space.[67,68] The results showed that a spinal injection of 0.5 mg/kg bupivacaine produces approximately 55 minutes of spinal anesthesia in these broiler chickens, which is much longer than the 18 minutes of anesthesia provided by 2 mg/kg lidocaine. The safe use of transdermal patches and creams, epidural infusions, spinal blocks, and IV blocks have not yet been reported in birds.

1.2.4. Other Drugs

Gabapentin is an analog of the neurotransmitter caminobutyric acid (GABA) that binds to voltage-gated calcium channels acting presynaptically to decrease the release of excitatory neurotransmitters. The pharmacokinetics of gabapentin have been evaluated in several species. In these studies a compounded formulation was used to avoid potential risks associated with xylitol in the commercially available oral suspension of gabapentin for humans. In great horned owls (*Bubo virginianus*) following a oral dose administration of 11 mg/kg PO, the Cmax, Tmax and elimination half-life were 6.17 ug/mL, 51.43 min and 4.4 hours, respectively. In this study, plasma gabapentin concentrations were maintained above 2 ug/mL for 9.6 hours, suggesting that gabapentin administered orally every 8 hours may be appropriate in great horned owls.[69] In Hispaniolan Amazon parrots (*Amazona ventralis*), oral bioavailability was estimated to be 80-89%.[70] Two and three birds respectively received 10 and 30 mg/kg PO, with a Cmax, Tmax and elimination half-life of 7.29 and 16.08 ug/dL, 2 and 2.5 hours, and 5.41 and 3.74 respectively. Mild transient sedation was noted after intravenous injection of gabapentin that resolved after 20–30 min. From that study the authors concluded that 15 mg/kg every 8 hours would be a starting point for oral dosing in Hispaniolan Amazon parrots based on effective plasma concentrations reported for human patients; however, additional studies were considered necessary. In a more recent study Caribbean flamingos (*Phoenicopterus ruber ruber*) following administration of a 15 mg/kg and 25 mg/kg dose, the Cmax, Tmax and elimination half-life 13.23 and 24.48 ug/mL, 0.50 and 0.56 hours, and 3.39 and 4.46 hours respectively.[71] Based on the results of this study, gabapentin dosed at 25 mg/kg orally in most Caribbean flamingos is likely to maintain plasma concentrations above the therapeutic range established for humans for approximately 12 hour; however, dosages as high as 37 mg/kg orally every 12 hour may be needed in some individuals.

Amantadine is an antiviral medication that increases dopamine in the central nervous system and weakly inhibits the NMDA receptors that contribute to chronic pain.[72] Amantadine is synergistic with NSAIDs, opioids, and gabapentin/pregabalin. In humans, amantadine has been shown to decrease opioid use in post-operative patients, decrease pain in postherpetic neuralgia, and aid in neuropathic analgesia.[73,74] Amantadine orally administered to dogs at 3-5 mg/kg every 24 hours in conjunction with meloxicam showed significant improvement in management of osteoarthritis pain refractory to meloxicam alone in dogs.[75] In orange-winged Amazon parrots, the single and multiple dose pharmacokinetic profile was evaluated following 10 mg/kg PO once and 5 mg/kg PO q24h for 7 days.[76] The heart rate, respiratory rate, behavior, and urofeces were monitored. Mean ± SD maximum plasma concentration, time to maximum plasma concentration, half-life, and area under the curve from the last dose to infinity were 1174 ± 186 ng/mL, 3.75 hours, 23.2 hours, and 38.6 h•µg/mL, respectively, for the single dose study; and 1185 ± 270 ng/mL, 3.0 hours, 21.5 hours, and 26.3 h•µg/mL, respectively, for the multiple dose study at steady-state. No obvious adverse effects were observed. Based on these results, once-daily administration of amantadine at 5 mg/kg orally in orange-winged Amazon parrots is able to maintain above target plasma concentration with weak accumulation. For long term administration, especially for other species, it is recommended to check plasma concentrations over time. Further studies evaluating safety and efficacy of amantadine in orange-winged Amazon parrots and other avian species is warranted. It is also important to remember that the use of amantadine in chickens, ducks, and turkeys is prohibited in the United States of America by the Federal Drug Administration.

Cannabinoids have been evaluated in other species for its multiple effects including pain relief and anti-inflammatory properties, making it an attractive therapeutic option in birds with chronic pain. Cannabidiol (CBD) is the primary non-psychoactive cannabinoid derived from the cannabis plant (*Cannabis sativa*) that can interact with the endocannabinoid system, but other cannabionoids like cannabidiolic acid (CBDA) have gained more recent interest. The single dose pharmacokinetics of a CBD oral formulation (50 mg/mL; Canna Companion, Mill Creek, WA, USA) have been evaluated in Hispaniolan Amazon parrots.[77] The C_{max} were 213 and 562 ng/mL at 0.5 and 4 hours, respectively with an elimination half-life elimination of 1.28 hours for the 120 mg/kg dose. The highly variable results and short half-life of the drug in Hispaniolan Amazon parrots, even at high doses, suggests that this formulation would have to be administered at least 3-4 times daily. More recently a commercial hemp (*Cannabis sativa*) extract (100 mg/mL; Ellevet, South Portland, ME, USA) that contains cannabidiol (CBD), cannabidiolic acid (CBDA), Δ -tetrahydrocannabinol 9 (THC), tetrahydrocannabinolic acid (THCA), cannabigerol (CBG), cannabigerolic acid (CBGA) and cannabichromene (45.2, 49, 1.86, 0.56, 0.69. 1.21, 1.53 mg/mL respectively) was evaluated in orange-winged Amazon parrots in a single and multiple dose pharmacokinetic and safety study at a dose of 30/32.5 mg/kg PO of CBD/CBDA.[78] The half-life in the multiple dose study was 8.6 and 6.3 hours for CBD and CBDA respectively. Interesting differences in the metabolism of some of the cannabinoids when compared to other species studied were found. The results from that study were encouraging for a q 12 hour administration, based on plasma concentrations and limited safety data in the 7 days multiple dose study. Evaluation of other formulations and also in other species is recommended to further characterize the analgesic potential of CBD, CBDA and other cannabinoids in birds.

1.1.1. Dietary Supplements

Omega-3 polyunsaturated fatty acids like eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) have anti-inflammatory effects that might reduce the pain associated with osteoarthritis. The total EPA and DHA dosages are primary factors to consider with the omega-3 to omega-6 fatty acid ratio a lesser factor. Other omega-3 fatty acids (e.g., plant-based omega-3 fatty acid, α-linoleic acid) do not have similar effects. There have been no studies in birds evaluating the effects and optimal dosage of EPA and DHA in osteoarthritic pain.

Glucosamine, methylsulfonylmethane, and chondroitin sulfate may have benefits in treating osteoarthritis through anti-inflammatory effects, but there are no studies in birds evaluating dosages and potential adverse effects.

Polysulfated glycosaminoglycans (PSGAGs) have also been used anecdotally in the management of degenerative joint disease in birds, but fatal coagulopathies following IM doses from 0.5 mg/kg to 100 mg/kg in different avian species (one coraciiform, two raptors, and one psittacine) have been reported.[79] A recent in vitro study evaluating the hypocoagulability effects the same specific formulation (100 mg/mL; Adequan, Elanco, Luitpold Animal Health, Greenfield, USA) previously associated with adverse effects, paired with the anecdotal reports of one of the authors of use in a wide range of species over 20 years, supports its safe and effective use in birds at 1 mg/kg SQ once weekly for four doses, followed by once every other week for two doses, then once monthly until the problem is resolved, or chronically at this dose for long-term maintenance.[80]

1.1.2. Physical Rehabilitation

Treatment for pain in humans and companion mammal species may include physical modalities, manual therapy, and therapeutic exercise. The application of adjunctive therapy should be considered for acute and chronic pain in birds, although evidence-based information is not yet available for birds. Therapeutic exercises, such as static weight-bearing, can be generally used in the acute phase of an injury with gradual progression of difficulty as healing occurs. Physical modalities such as thermotherapy and low-level laser are used to diminish pain. Thermotherapy can have analgesic effects after the inflammatory effect has subsided. Low-level laser therapy has been shown to decrease indicators of neuropathic pain. Manual physical therapeutic techniques for joint mobilization can decrease pain but trigger point-pressure techniques might induce central sensitization.[81]

1.1.1. Supportive Care

Cold compress during acute injury can reduce swelling and provide analgesia. Cold compress generally needs to be in place for 15 to 20 minutes to be effective. Warm compress is generally more comfortable after the acute phase has passed, and can aid tissue relaxation and be used as a precursor to massage or stretching. Warm compress generally needs to be in place for 10 to 15 minutes.[79,81]

The environment may affect pain, and stress and anxiety can have a modulatory effect. The bird should be in an environment where it is emotionally and physically comfortable. Human presence or absence, decreasing light and noise, and separating other animals from the bird can reduce stress and anxiety during the pain period. When handling and moving an animal avoid painful areas (surgical/trauma site, osteoarthritic joints, etc.), even when the animal is anesthetized or sedated, to avoid inflicting a painful stimulus that can begin a new pain cascade.[81]

2. GENERAL ANESTHESIA

2.1. THE RISK OF DEATH

One study in small animals has provided some data regarding peri-anesthetic mortality in birds.[82] In this study, the peri-anesthetic mortality rate was found to be 3.94 % in parrots (N=127, 95%CI = 1.29–8.95%) which is approximately 20 times higher than those found in dogs and cats. The mortality rate was higher for budgerigars with 16.33% (N=49, 95% CI 7.32-29.66), and lower for "other birds" with 1.76% (N=284, 95% CI= 0.57-4.06%). This study did not provide further information regarding the species included as "other birds', the health status of the patient, anesthetic protocols or degree of monitoring and support during the procedure. There are certainly many additional challenges when anesthetizing avian patients, for example, difficulty in obtaining pre-operative blood work, securing airways or IV access in very small patients, or difficulty monitoring blood pressures. Nevertheless, this study on anesthetic-associated deaths in these animals serves as a warning when choosing an anesthetic protocol and evaluating safety and efficacy. A more recent study looked at the avian mortality rate (N=1820) based on ASA status,[83] finding that patients assigned an ASA grade of 1 and 2 had a rate of 0% (95%CI= 0.0-0.2) and 0.6% (CI 0.2-1.3) while patients assigned grades 3, 4, and 5 had a rate of 5.9% (CI 4.3-8.0), 18.8% (CI 13.4-25.7) and 50.0% (CI 29.9 70.1) respectively. Investigation into timing of death showed that the majority of patients died following cessation of general anesthesia (81.48%), with the highest mortality rate occurring between 0 and 3 hours postgeneral anesthesia.

2.2. PREANESTHETIC PREPARATION

Before anesthesia, the patient should be evaluated. The preanesthetic evaluation should include a history and a physical examination. The laboratory evaluation should include a complete blood count and a biochemistry panel. If any abnormalities are detected (e.g. dehydration), should be corrected with appropriate therapy (e.g. fluid therapy) before the procedure. The fasting period is variable, ranging from 2-4 hours in most psittacines to 24 hours in most raptors species.

2.3. PREMEDICATION

Premedication prior to induction facilitates handling and decreases the anesthetic and analgesic drugs required during the procedure. Most commonly, a combination of an opioid drug (e.g. butorphanol in pisttacines or hydromorphone in raptors) and a benzodiazepine (e.g. midazolam) is used. Benzodiazepines (e.g. diazepam, midazolam) are increasingly used in birds and produce sedation, hypnosis, anxiolysis, presumed anterograde amnesia, centrally mediated muscle relaxation and anti-convulsion. Midazolam (0.2-2 mg/kg) is preferred over diazepam because can be administered intramuscularly or intranasally.[84] Parasympatholytics (e.g. atropine, glycopyrrolate) are not routinely administered to birds because of the belief that increase the viscosity of the respiratory tract secretions increasing the risk of airway or endotracheal tube obstruction.[85-87]

2.4. INJECTABLE AGENTS

Injectable anesthetics, such as propofol or medetomidine–ketamine or xylazine-ketamine (followed by reversal with atipamezole or yohimbine), may be useful for remote field locations where commercial shipment of hazardous gases and anesthetic agents is logistically complicated or prohibited or for use in large birds for which mask induction may be challenging (e.g., ratites). Propofol[88-97] and ketamine-medetomidine or xylazine combinations[93,94,98-102] have been evaluated in several avian species. Injectable anesthetics might be preferred also in surgeries involving the beak, mouth, glottis, coelomic cavity, respiratory system, or pneumatic bones. The need for ventilatory support, the difficulty to maintain a constant plane of anesthesia and the potential for excitatory and/or prolonged recoveries limit their use in companion birds as a sole agent.

2.5. INHALATION AGENTS

Inhalation anesthesia is more commonly used than injectable anesthesia in clinical avian practice. Isoflurane is currently the anesthetic agent of choice, although sevoflurane is also an excellent, more expensive, option. Both, isoflurane and sevoflurane, are considered dose dependent respiratory depressants. Isoflurane results in rapid inductions, allows rapid alterations in anesthetic plane, has a small margin between respiratory and cardiac arrest, and provides rapid and smooth recoveries. Sevoflurane, due to lower solubility than isoflurane, allows for faster induction and recovery, as well as more rapid changes in anesthetic depth, but this difference has proven to be small or not existent in some studies in different avian species. Sevoflurane also appears to be less pungent to the airways in comparison to isoflurane. The cardiopulmonary differences between isoflurane and sevoflurane have not been fully evaluated in psittacine birds.[103] In bald eagles, isoflurane administration resulted in tachycardia, hypertension, and more arrhythmias, compared with sevoflurane,[104] and in red-tailed hawks there was a significantly lower spontaneous respiratory rate in isoflurane when compared with sevoflurane.[105] The minimum anesthetic concentrations (MAC) of sevoflurane and isoflurane have been determined only in a few species. Reported MAC values for isoflurane include thick-billed parrots (1.07 ± 0.1%),[106] chickens (1.24 ± 0.05%),[107] sandhill cranes (1.34 ± 0.14%),[108] Pekin ducks (1.30 ± 0.23%)[109] and cockatoos (1.44 ± 0.07%).[110] Reported MAC values for sevoflurane include thick-billed parrots (2.35%),[111] and chickens 2.21 ± 0.32%.[112] During induction, the bird is manually restrained until immobilized and then placed in sternal or lateral recumbency position on a padded surface. Birds are commonly induced at 3 % with isoflurane or 5 % with sevoflurane, and usually takes 1-3 minutes. Alternatively, can be induced at lower concentrations, taking usually 1-5 minutes. Intubation is performed after the patient is induced into a medium plane of anesthesia, and care should be taken to prevent tracheal trauma. The endotracheal tube is secured with umbilical cord tape or regular tape. Birds are commonly maintained at 1-2.5% isoflurane and 3-4 % sevoflurane for most procedures. An oxygen flow rate of 150 to 200 mL/kg/min is reported, but 1 L/min in most medium and large birds (approximately >300 g) is commonly used.

2.6. INHALATION EQUIPMENT

Non-rebreathing circuits, such as the Bain circuit, are recommended for avian anesthesia. The advantages of these circuits include decreased resistance to breathing by the patient and rapid responses to changes in the vaporizer setting. A 0.5- to 1.0-L bag is used for most birds.

Commercially available masks designed for dogs and cats can be used for induction of avian anesthesia. A properly sized mask allows the entire head and beak to be inserted while minimizing dead space. Masks can be modified to accommodate the widely diverse anatomic variations of avian species. For example, for budgerigars and finches, a cut-down 35- or 60-mL syringe case can be used, with a hole drilled at the closed end to facilitate insertion of the adapter piece for the circuit. At the other extreme, a 1- to 2-liter soft drink bottle can be cut down to facilitate masking a large toucan.

Endotracheal uncuffed tubes are used in birds, since the use of cuff should be avoided because of the presence of complete tracheal rings. The endotracheal tube used in birds varies from 1.0-6 mm diameter. Cole tubes provide a good seal of the glottis. While endotracheal tubes for birds weighing less than 100 g are commercially available (e.g. 1.0 mm diameter), red rubber feeding tubes are a good alternative. Great caution should be exercised with endotracheal tubes smaller than 2.0 mm diameter since they can get obstructed easily. The increase in airway resistance is of greatest importance in very small patients since the resistance varies with the fourth power of the radius. For example, decreasing the radius from 4 to 3 mm would increase the resistance threefold, whereas a decrease from 3 to 2 mm and 2 to 1 mm would increase resistance approximately 5 and 15 times respectively.

2.7. MONITORING AND SUPPORT

Due to the rapid rate at which birds may decompensate under anesthesia, monitoring is the most important aspect of avian anesthesia. This means that having one person dedicated to consistent, hands-on monitoring and ensuring that the anesthetist has a clear view of the patient are critically important.

2.7.1. Central Nervous System

The loss of muscle tone in the legs or wings can be useful for assessing transition from a light to a medium plane of anesthesia. The withdraw reflexes (i.e., toe pinch) are lost when the bird is in a medium (surgical) plane of anesthesia. The palpebral reflexes are usually lost by a medium plane of anesthesia, but corneal reflexes persist until the deep plane of anesthesia. The heart and respiratory rates increase when the bird is experiencing pain or when the depth of anesthesia is too low; conversely, these rates may decrease in the deep plane of anesthesia, signaling the need to adjust the flow of anesthetic gases accordingly.

2.7.2. Cardiovascular System

The stethoscope can be used to monitor heart rate and rhythm, and the use of an esophageal probe can facilitate this job. The esophageal probe should be placed into the thoracic esophagus bypassing the crop with digital manipulation. The ultrasonic Doppler flow detector can be used for the same function. The sensor is commonly placed over the cranial tibial artery, palpable on the cranial aspect of the hock joint; the superficial ulnar artery, palpable on the ventral surface of the elbow joint; the deep radial artery, palpable on the ventral surface of the distal radius near the carpal joint or the palatal artery in the dorsal oropharynx. The ECG can be used as well for monitoring of the heart rate and rhythm, and the electrodes are positioned at the patagium and inguinal skin web locations using flattened alligator clips, or small hypodermic needles are passed through the skin and attached to alligator clips.

Blood pressure may be subjectively assessed by palpating the pulse at the median ulnar or medial metatarsal artery. Direct arterial pressure monitoring can also be performed. The most common sites include the brachial and carotid arteries. Indirect blood pressure monitoring is performed using a Doppler ultrasonic probe to detect arterial flow, a pressure cuff for occlusion and a sphygmomanometer to measure pressure. Based on the studies in avian species, indirect blood pressure measurements do not have good agreement with direct arterial measurements[113,114] and are very variable between repeated measurements when cuffs are applied multiple times to the same limb.[115] Thus the results during an anesthetic procedure should be considered in terms of a trend, more than for evaluation of a single measurement. Indirect mean arterial pressures reported in Hispaniolan Amazon in the wing were 140 ± 25mmHg, while in the leg were 145 ± 28 mmHg, with a cuff width of 30-40% of the diameter.[60] In red-tailed hawks, mean arterial pressures were 155 ± 27 mmHg from both wing and leg with a cuff width of 40-50% of the diameter. Oscillometric devices have demonstrated high failure rates and unreliable measurements in avian patients.[114]

Fluid therapy is a critical component of avian anesthesia, particularly for procedures lasting longer than 20 minutes. Ideally, an intravenous or intraosseous (IO) catheter is placed to facilitate fluid administration at 10 mL/kg/h; however, subcutaneous fluid administration may be adequate for routine or relatively short procedures in otherwise stable patients. Dobutamine CRI at 5-15 ug/kg/min and dopamine CRI at 5-10 ug/kg/min have shown to help correcting severe hypotension in Hispaniolan Amazon parrots caused by anesthesia maintained with 2.5% isoflurane.[116]

2.7.3. Respiratory System

Respiratory rate and character should be closely monitored throughout anesthesia. Ventilation is assessed by observing the frequency and range of sternal motion and reservoir bag movement. A change in respiratory pattern involving increased respiratory effort may indicate obstruction of the endotracheal tube, a particular concern in very small patients. Normal ventilation is associated with little to no respiratory tract noise. In the few species that have been studied it appears that hypercapnia is associated with cardiovascular depression in birds and so the support of ventilation may be more routine in avian anesthesia.

The capnograph is used to monitor end-tidal carbon dioxide ($PETCO_2$) and provides information regarding ventilatory status. There is good correlation between $PETCO_2$ and $PaCO_2$, with $PETCO_2$ slightly exceeding PaCO2.[117,118] An $ETCO_2$ of 30 to 45 mm Hg indicates adequate ventilation during inhalation anesthesia in most birds and approximates a normal physiologic range of 25 to 40 mmHg for $PaCO_2$ for awake birds. Intermittent positive-pressure ventilation (IPPV) is recommended at 2 breaths/min in spontaneously ventilating birds, and 10-20 breaths/min in apneic birds. A maximum pressure of 10-24 cm H_2O is recommended. If used in combination with a capnograph, IPPV rate should be adjusted to maintain $ETCO_2$ between 30-45 mmHg.

Pulse oximetry has not been validated in birds.[119] The absorption characteristics of oxygenated and deoxygenated avian and human hemoglobin are different resulting in underestimation of hemoglobin saturation. [119 118 117 116] However, these monitors are often used reliably to provide a trend in the oxygen saturation and a pulse rate.[119]

2.7.4. Body Temperature

The birds core body temperature should be monitored throughout anesthesia via a cloacal or esophageal temperature probe. Avian esophageal and cloacal temperatures are well correlated in studies that compared the two sites; however, cloacal temperature probes are prone to being dislodged or recording a cooler temperature if they are not secured appropriately within the cloaca.[120] Thus, the temperature is best monitored using esophageal probes, and it should be maintained between 40 and 41.2 °C (104-106 ºF).[120]

Thermal support is necessary to prevent hypothermia in birds under general anesthesia. Radiant heat sources, water blankets, fluid warmer and forced air warmer systems (e.g., Bair Hugger Warming System) are routinely use to maintain core body temperature. The most effective method for maintaining the temperature has been found to be the forced-air warming device while covering the bird with a transparent drape, when compared to water blanket or radiant heat source with a drape.[121] The type of anesthetic delivery system, with or without heated air, does not appear to affect the core body temperature.[122]

2.8. ANESTHETIC EMERGENCIES

Emergencies during anesthesia should be expected and planned for. Careful monitoring and support during anesthesia of the nervous, cardiovascular and respiratory function as well as body temperature. It is recommended that emergency drug dosages are calculated and 1–2 doses be predrawn before induction and that the emergency equipment (e.g.., endotracheal tubes, oxygen, IV catheters and materials for securing them, ventilatory support, and emergency drugs) is prepared and accessible for use.

2.8.1. Bradypnea and Hypoventilation

During anesthetic procedures, some degree of respiratory depression should be expected, which will increase over time. Because of that, tracheal intubation with intermittent positive-pressure ventilation (IPPV) is recommended at 2 breaths/min in spontaneously ventilating birds, and 10-20 breaths/min in birds with significant bradypnea. If used in combination with a capnograph, IPPV rate should be adjusted to maintain $ETCO_2$ between 30-45 mmHg. The anesthetist should also assess the frequency and range of sternal motion and reservoir bag movement. A change in respiratory pattern involving increased respiratory effort may indicate obstruction of the endotracheal tube, a particular concern in very small patients. If and obstruction is suspected, repositioning of the bird with the neck extended and/or suction through the endotracheal tube is recommended to resolve any possible blockage. Normal ventilation is associated with little to no respiratory tract noise.

2.8.2. Respiratory Arrest

If respiratory arrest occurs, tracheal intubation and intermittent positive-pressure ventilation (IPPV) is required at 10-20 breaths/min in birds. The level of gas anesthesia may be reduced, if indicated, but not necessarily discontinued. If used in combination with a capnograph, IPPV rate should be adjusted to maintain $ETCO_2$ between 30-45 mmHg. The ventilatory support should be continued throughout the anesthetic procedure or until spontaneous ventilation resumes. If corrected immediately, the prognosis for respiratory arrest is good. The use of doxapram as a treatment for apnea is controversial and is not recommended. [123]

2.8.3. Bradycardia and Hypotension

Bradycardia and hypotension (defined as systolic BP less than 90 mm Hg) develop secondary to several conditions during anesthesia, including cardiovascular depression caused by the anesthetic agent, hypothermia, vagal response due to severe pain, hypovolemia related to dehydration or blood loss. The anesthetist must assess whether it represents a true emergency (e.g continues blood loss during a procedure) or is a transient response (e.g. vagal response due to pain during fracture manipulation). The level of inhalant anesthesia should be reduced if bradycardia and hypotension are thought to be caused by a high anesthetic plane. The administration of IV fluids is indicated if the bradycardia and hypotension are attributed to hypovolemia or for immediate correction of hypotensive state. The anticholinergic agents such as atropine (fast acting, short lasting) or glycopyrrolate (slow acting, long lasting) can be used to increase the heart but might result in sinus tachycardia and increase myocardial work and oxygen consumption. an Intravenous or intraosseous (IO) administration of warmed crystalloid fluids at10 mL/kg/h or 10-15 mL as a bolus over 5-10 min is indicated for hypovolemic birds and can be repeated if necessary to increase systolic BP to more than 90 mm Hg. Synthetic colloid solutions (i.e., hetastarch [Abbott Laboratories, North Chicago, IL], 5 mL/kg IV over 5-10 min) may be used in hypoproteinemic birds or birds that have suffered acute blood loss during surgery. Blood transfusions may also be indicated for acute blood loss at a rate of 5 mL/kg/h. Dobutamine administered at 5-15 ug/kg/min and dopamine at 5-10 ug/kg/min have shown to help correcting severe hypotension in Hispaniolan Amazon parrots caused by anesthesia maintained with 2.5% isoflurane.[116]

2.8.4. Cardiac Arrest

If cardiac arrest develops, continue ventilatory support as explained above in the respiratory arrest section. The solid keel interferes with effective cardiac massage, although in the event of cardiac arrest, attempts should be made to compress the sternum at a rate of 60 to 100 compressions/min. Emergency drugs, including epinephrine (1:1000; 0.01 mg/kg IV, IO, or a double dosage IT), atropine (0.04 mg/kg IV, IO, or a double dosage IT) should be administered for 2-3 rounds. The use of vasopressin at a dosage of 0.8 U/kg IV, IO or a double dosage IT in addition or instead of epinephrine have been suggested. [124] The anesthetist should assess the efforts during cardiopulmonary resuscitation to see if there is spontaneous pulse. These efforts should be continued for up to five minutes. The prognosis for recovery is poor for avian patients that go into cardiac arrest, and although rare, some birds might recover.

2.9. RECOVERY

During recovery, the bird is gently restrained in a towel and extubated once it starts to resist the presence of the tube, but oxygen is administered via an open mask through the recovery from anesthesia. A quiet, warm (approximately 25°C; 77°F), and light-reduced environment should be selected; which allows for visual control of the patient.

REFERENCES

1. Gentle MJ, Hunter LN. Physiological and behavioural responses associated with feather removal in Gallus gallus var domesticus. *Res Vet Sci* 1991;50:95-101.

2. Mazor-Thomas JE, Mann PE, Karas AZ, et al. Pain-suppressed behaviors in the red-tailed hawk *Applied animal behaviour science* 2014;152:83-91.

3. Mikoni NA, Sanchez-Migallon Guzman D, Beaufrere H, et al. Carrageenan-induced inflammation elicits behavioral changes in cockatiels (Nymphicus hollandicus) for potential pain scale development. *Am J Vet Res* 2023;84:1-11.

4. Mikoni NA, Guzman DS, Paul-Murphy J. Pain Recognition and Assessment in Birds. *Vet Clin North Am Exot Anim Pract* 2023;26:65-81.

5. Desmarchelier M, Troncy E, Beauchamp G, et al. Evaluation of a fracture pain model in domestic pigeons (Columba livia). *Am J Vet Res* 2012;73:353-360.

6. Sanchez-Migallon Guzman D, Flammer K, Paul-Murphy JR, et al. Pharmacokinetics of butorphanol after intravenous, intramuscular, and oral administration in Hispaniolan Amazon parrots (Amazona ventralis). *J Avian Med Surg* 2011;25:185-191.

7. Paul-Murphy J, Brunson DB, Miletic V. Analgesic effects of butorphanol and buprenorphine in conscious African grey parrots (Psittacus erithacus erithacus and Psittacus erithacus timneh). *Am J Vet Res* 1999;60 1218-1221.
8. Guzman DS, Flammer K, Paul-Murphy JR, et al. Pharmacokinetics of butorphanol after intravenous, intramuscular, and oral administration in Hispaniolan Amazon parrots (Amazona ventralis). *J Avian Med Surg* 2011;25:185-191.
9. Sladky K, Krugner-Higby L, Meek-Walker E, et al. Serum concentrations and analgesic effects of liposome-encapsulated and standard butorphanol tartrate in parrots. *Am J Vet Res* 2006;67:775-781.
10. Klaphake E, Schumacher J, Greenacre C, et al. Comparative anesthetic and cardiopulmonary effects of pre- versus postoperative butorphanol administration in Hispaniolan amazon parrots (Amazona ventralis) anesthetized with sevoflurane. *J Avian Med Surg* 2006;20.
11. Guzman DS, Drazenovich TL, KuKanich B, et al. Evaluation of thermal antinociceptive effects and pharmacokinetics after intramuscular administration of butorphanol tartrate to American kestrels (Falco sparverius). *Am J Vet Res* 2014;75:11-18.
12. Singh PM, Johnson CB, Gartrell B, et al. Analgesic effects of morphine and butorphanol in broiler chickens. *Vet Anaesth Analg* 2017;44:538-545.
13. Guzman DS, Drazenovich TL, Olsen GH, et al. Evaluation of thermal antinociceptive effects after intramuscular administration of hydromorphone hydrochloride to American kestrels (Falco sparverius). *Am J Vet Res* 2013;74:817-822.
14. Houck EL, Guzman DS, Beaufrere H, et al. Evaluation of the thermal antinociceptive effects and pharmacokinetics of hydromorphone hydrochloride after intramuscular administration to cockatiels (Nymphicus hollandicus). *Am J Vet Res* 2018;79:820-827.
15. Durant A, Guzman DS-M, Beaufrère H. Thermal antinociceptive effects after intramuscular administration of high doses of hydromorphone in cockatiels (Nymphicus hollandicus). *Am J Vet Res* 2025;1:1-6.
16. Hoppes S, Flammer K, Hoersch K, et al. Disposition and analgesic effects of fentanyl in the umbrella cockatoo (Cacatua alba). *J Avian Med Surg* 2003;17:124-130.
17. Pavez JC, Hawkins MG, Pascoe PJ, et al. Effect of fentanyl target-controlled infusions on isoflurane minimum anaesthetic concentration and cardiovascular function in red-tailed hawks (Buteo jamaicensis). *Veterinary Anaesthesia and Analgesia* 2011;38:344-351.
18. Hawkins MG, Pascoe PJ, DiMaio Knych HK, et al. Effects of three fentanyl plasma concentrations on the minimum alveolar concentration of isoflurane in Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2018;79:600-605.
19. Delaski KM, Gehring R, Heffron BT, et al. Plasma concentrations of fentanyl achieved with transdermal application in chickens. *J Avian Med Surg* 2017;31:6-15.
20. Paul-Murphy JR, Brunson DB, Miletic V. Analgesic effects of butorphanol and buprenorphine in conscious African grey parrots (Psittacus erithacus erithacus and Psittacus erithacus timneh). *Am J Vet Res* 1999;60:1218-1221.
21. Paul-Murphy J, Hess JC, Fialkowski JP. Pharmacokinetic properties of a single intramuscular dose of buprenorphine in African Grey parrots (Psittacus erithacus erithacus). *J Avian Med Surg* 2004;18:224-228.
22. Ceulemans S, Sanchez-Migallon Guzman D, Drazenovich T, et al. Evaluation of the thermal antinociceptive effects after administration of intramuscular buprenorphine hydrochloride in American kestrels (Falco sparverius). *Am J Vet Res* 2014;In press.
23. Guzman DS, Houck EL, Knych HKD, et al. Evaluation of the thermal antinociceptive effects and pharmacokinetics after intramuscular administration of buprenorphine hydrochloride to cockatiels (Nymphicus hollandicus). *Am J Vet Res* 2018;79:1239-1245.
24. Guzman DS, Ceulemans SM, Beaufrere H, et al. Evaluation of the Thermal Antinociceptive Effects of a Sustained-Release Buprenorphine Formulation After Intramuscular Administration to American kestrels (Falco sparverius). *J Avian Med Surg* 2018;32:1-7.
25. Guzman DS, Knych HK, Olsen GH, et al. Pharmacokinetics of a sustained release formulation of buprenorphine after intramuscular and subcutaneous administration to American kestrels (Falco sparverius). *J Avian Med Surg* 2017;31:102-107.
26. Gleeson MD, Guzman DS, Knych HK, et al. Pharmacokinetics of a concentrated buprenorphine formulation in red-tailed hawks (Buteo jamaicensis). *Am J Vet Res* 2018;79:13-20.
27. Guzman DS-M, Douglas J, Beaufrère H, et al. Evaluation of thermal antinociceptive effects after intramuscular administration of concentrated buprenorphine to orange-winged Amazon parrots (Amazona amazonica). *Unpublished* 2020.
28. Sanchez-Migallon Guzman D, Souza MJ, Braun JM, et al. Antinociceptive effects after oral administration of tramadol hydrochloride in Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2012;73:1148-1152.

29. Souza MJ, Sanchez-Migallon Guzman D, Paul-Murphy JR, et al. Pharmacokinetics after oral and intravenous administration of a single dose of tramadol hydrochloride to Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2012;73:1142-1147.
30. Souza MJ, Martin-Jimenez T, Jones MP, et al. Pharmacokinetics of intravenous and oral tramadol in the bald eagle (Haliaeetus leucocephalus). *J Avian Med Surg* 2009;23:247-252.
31. Guzman DS, Drazenovich TL, Olsen GH, et al. Evaluation of thermal antinociceptive effects after oral administration of tramadol hydrochloride to American kestrels (Falco sparverius). *Am J Vet Res* 2014;75:117-123.
32. Bailey RS, Sheldon JD, Allender MC, et al. Analgesic Efficacy of Tramadol Compared With Meloxicam in Ducks (Cairina moschata domestica) Evaluated by Ground-Reactive Forces. *J Avian Med Surg* 2019;33:133-140.
33. Cole GA, Paul-Murphy J, Krugner-Higby L, et al. Analgesic effects of intramuscular administration of meloxicam in Hispaniolan parrots (Amazona ventralis) with experimentally induced arthritis. *Am J Vet Res* 2009;70:1471-1476.
34. Desmarchelier M, Troncy E, Fitzgerald G, et al. Analgesic effects of meloxicam administration on postoperative orthopedic pain in domestic pigeons (Columba livia). *Am J Vet Res* 2012;73:361-367.
35. Baert K, De Backer P. Comparative pharmacokinetics of three non-steroidal anti-inflammatory drugs in five bird species. *Comp Biochem Physiol C Toxicol Pharmacol* 2003;134:25-33.
36. Wilson G, Hernandez-Divers S, Budsberg S, et al. Pharmacokinetics and use of meloxican in psittacine birds. Proc Assoc Avian Vet 2004;7-9.
37. Naidoo V, Wolter K, Cromarty AD, et al. The pharmacokinetics of meloxicam in vultures. *J Vet Pharmacol Ther* 2008;31:128-134.
38. Montesinos A, Ardiaca M, Gilabert JA, et al. Pharmacokinetics of meloxicam after intravenous, intramuscular and oral administration of a single dose to African grey parrots (Psittacus erithacus). *J Vet Pharmacol Ther* 2016.
39. Lacasse C, Gamble KC, Boothe DM. Pharmacokinetics of a single dose of intravenous and oral meloxicam in red-tailed hawks (Buteo jamaicensis) and great horned owls (Bubo virginianus). *J Avian Med Surg* 2013;27:204-210.
40. Molter CM, Court MH, Cole GA, et al. Pharmacokinetics of meloxicam after intravenous, intramuscular, and oral administration of a single dose to Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2013;74:375-380.
41. Dhondt L, Devreese M, Croubels S, et al. Comparative population pharmacokinetics and absolute oral bioavailability of COX-2 selective inhibitors celecoxib, mavacoxib and meloxicam in cockatiels (Nymphicus hollandicus). *Sci Rep* 2017;7:12043.
42. Pereira M, Werther K. Evaluation of the renal effects of flunixin meglumine, ketoprofen and meloxicam in budgerigars (Melopsittacus undulatus). *Vet Rec* 2007.
43. Montesinos A, Ardiaca M, Juan-Selles C, et al. Evaluation of the renal effects of meloxicam in African Grey parrots (Psittacus erithacus erithacus). Proc European Ass Avian Vet 2009;171-176.
44. Dijkstra B, Guzman DS, Gustavsen K, et al. Renal, gastrointestinal, and hemostatic effects of oral administration of meloxicam to Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2015;76:308-317.
45. Sinclair KM, Church ME, Farver TB, et al. Effects of meloxicam on hematologic and plasma biochemical analysis variables and results of histologic examination of tissue specimens of Japanese quail (Coturnix japonica). *Am J Vet Res* 2012;73:1720-1727.
46. Summa NM, Guzman DS, Larrat S, et al. Evaluation of High Dosages of Oral Meloxicam in American Kestrels (Falco sparverius). *J Avian Med Surg* 2017;31:108-116.
47. Montesinos A, Encinas T, Ardiaca M, et al. Pharmacokinetics of meloxicam during multiple oral or intramuscular dose administration to African grey parrots (Psittacus erithacus). *Am J Vet Res* 2019;80:201-207.
48. Oaks JL, Gilbert M, Virani MZ, et al. Diclofenac residues as the cause of vulture population decline in Pakistan. *Nature* 2004;427:630-633.
49. Naidoo V, Wolter K, Cromarty D, et al. Toxicity of non-steroidal anti-inflammatory drugs to Gyps vultures: a new threat from ketoprofen. *Biol Lett* 2010;6:339-341.
50. Swarup D, Patra RC, Prakash V, et al. Safety of meloxicam to critically endangered Gyps vultures and other scavenging birds in India. *Animal Conservation* 2007;10:192-198.
51. Cuthbert R, Parry-Jones J, Green RE, et al. NSAIDs and scavenging birds: potential impacts beyond Asia's critically endangered vultures. *Biol Lett* 2007;3:90-93.
52. Guzman DS, Court MH, Zhu Z, et al. Pharmacokinetics of a Sustained-release Formulation of Meloxicam After Subcutaneous Administration to Hispaniolan Amazon Parrots (Amazona ventralis). *J Avian Med Surg* 2017;31:219-224.
53. Sim RR, Cox SK. Pharmacokinetics of a Sustained-Release Formulation of Meloxicam after Subcutaneous Administration to American Flamingos (Phoenicopterus Ruber). *J Zoo Wildl Med* 2018;49:839-843.
54. Rodriguez P, Paul-Murphy JR, Knych HK, et al. Pharmacokinetics of grapiprant administered to red-tailed hawks (Buteo jamaicensis) after food was withheld for 24 hours. *Am J Vet Res* 2021;82:912-919.

55. Rodriguez P, Paul-Murphy JR, Knych HK, et al. Absorption of grapiprant in red-tailed hawks (Buteo jamaicensis) is decreased when administered with food. *Am J Vet Res* 2022;83.
56. Brandao J, da Cunha AF, Pypendop B, et al. Cardiovascular tolerance of intravenous lidocaine in broiler chickens (Gallus gallus domesticus) anesthetized with isoflurane. *Vet Anaesth Analg* 2014.
57. Da Cunha AF, Messenger KM, Stout RW, et al. Pharmacokinetics of lidocaine and its active metabolite monoethylglycinexylidide after a single intravenous administration in chickens (Gallus domesticus) anesthetized with isoflurane. *J Vet Pharmacol Ther* 2012;35:604-607.
58. Escobar A, Dzikiti BT, Thorogood JC, et al. Effects of two continuous infusion doses of lidocaine on isoflurane minimum anesthetic concentration in chickens. *Vet Anaesth Analg* 2023;50:91-97.
59. Hocking PM, Gentle MJ, Bernard R, et al. Evaluation of a protocol for determining the effectiveness of pretreatment with local analgesics for reducing experimentally induced articular pain in domestic fowl. *Res Vet Sci* 1997;63 263-267.
60. Glatz PC, Murphy LB, Preston AP. Analgesic therapy of beak-trimmed chickens. *Aust Vet J* 1992;69:18.
61. Machin KL, A. L. Plasma bupivicaine levels in mallard ducks (Anas platyrhynchos) following a single subcutaneous dose. Proc Am Assoc Zoo Vet 2001;159-163.
62. DiGeronimo PM, da Cunha AF, Pypendop B, et al. Cardiovascular tolerance of intravenous bupivacaine in broiler chickens (Gallus gallus domesticus) anesthetized with isoflurane. *Vet Anaesth Analg* 2017;44:287-294.
63. Brenner DJ, Larsen RS, Dickinson PJ, et al. Development of an avian brachial plexus nerve block technique for perioperative analgesia in mallard ducks (Anas platyrhynchos). *J Avian Med & Surg* 2010;24:24-34.
64. Figueiredo JP, Cruz ML, Mendes GM, et al. Assessment of brachial plexus blockade in chickens by an axillary approach. *Veterinary Anaesthesia and Analgesia* 2008;35:511-518.
65. da Cunha AF, Strain GM, Rademacher N, et al. Palpation- and ultrasound-guided brachial plexus blockade in Hispaniolan Amazon parrots (Amazona ventralis). *Veterinary Anaesthesia and Analgesia* 2013;40:1-7.
66. d'Ovidio D, Noviello E, Adami C. Nerve stimulator-guided sciatic-femoral nerve block in raptors undergoing surgical treatment of pododermatitis. *Vet Anaesth Analg* 2014.
67. Khamisabadi A, Kazemi-Darabadi S, Akbari G. Comparison of Anesthetic Efficacy of Lidocaine and Bupivacaine in Spinal Anesthesia in Chickens. *J Avian Med Surg* 2021;35:60-67.
68. Kazemi-Darabadi S, Akbari G, Shokrollahi S. Development and evaluation of a technique for spinal anaesthesia in broiler chickens. *N Z Vet J* 2019;67:241-248.
69. Yaw TJ, Zaffarano BA, Gall A, et al. Pharmacokinetic Properties of a Single Administration of Oral Gabapentin in the Great Horned Owl (Bubo Virginianus). *J Zoo Wildl Med* 2015;46:547-552.
70. Baine K, Jones MP, Cox S, et al. Pharmacokinetics of Compounded Intravenous and Oral Gabapentin in Hispaniolan Amazon Parrots (Amazona ventralis). *J Avian Med Surg* 2015;29:165-173.
71. Browning GR, Carpenter JW, Magnin GC, et al. Pharmacokinetics of Oral Gabapentin in Caribbean Flamingos (Phoenicopterus Ruber Ruber). *J Zoo Wildl Med* 2018;49:609-616.
72. KuKanich B. Outpatient oral analgesics in dogs and cats beyond nonsteroidal antiinflammatory drugs: an evidence-based approach. *Vet Clin North Am Small Anim Pract* 2013;43:1109-1125.
73. Bujak-Gizycka B, Kacka K, Suski M, et al. Beneficial effect of amantadine on postoperative pain reduction and consumption of morphine in patients subjected to elective spine surgery. *Pain Med* 2012;13:459-465.
74. Elmawgood AA, Rashwan S, Rashwan DJEJoA. Tourniquet-induced cardiovascular responses in anterior cruciate ligament reconstruction surgery under general anesthesia: Effect of preoperative oral amantadine. 2015;31:29-33.
75. Lascelles BD, Gaynor JS, Smith ES, et al. Amantadine in a multimodal analgesic regimen for alleviation of refractory osteoarthritis pain in dogs. *J Vet Intern Med* 2008;22:53-59.
76. Berg KJ, Sanchez-Migallon Guzman D, Knych HK, et al. Pharmacokinetics of amantadine after oral administration of single and multiple doses to orange-winged Amazon parrots (Amazona amazonica). *Am J Vet Res* 2020;81:651-655.
77. Carpenter JW, Tully Jr TN, Rockwell K, et al. Pharmacokinetics of Cannabidiol in the Hispaniolan Amazon Parrot (Amazona ventralis). 2022;36:121-127.
78. Sosa-Higareda M, Guzman DS, Knych H, et al. Twice-daily oral administration of a cannabidiol and cannabidiolic acid-rich hemp extract was well tolerated in orange-winged Amazon parrots (Amazona amazonica) and has a favorable pharmacokinetic profile. *Am J Vet Res* 2023:1-11.
79. Anderson K, Garner MM, Reed HH, et al. Hemorrhagic diathesis in avian species following intramuscular administration of polysulfated glycosaminoglycan. *J Zoo Wildl Med* 2013;44:93-99.
80. Wonn AM, Brooks MB, Hu H, et al. Hypocoagulability effect of Adequan in domestic chickens (Gallus gallus) and chilean flamingos (Phoenicopterus chilensis). *Journal of Zoo Wildlife Medicine* 2022;53:126-132.

81. Mathews K, Kronen PW, Lascelles D, et al. Guidelines for recognition, assessment and treatment of pain: WSAVA Global Pain Council members and co-authors of this document. *The Journal of small animal practice* 2014;55:E10-68.
82. Brodbelt DC, Blissitt KJ, Hammond RA, et al. The risk of death: the confidential enquiry into perioperative small animal fatalities. *Vet Anaesth Analg* 2008;35:365-373.
83. Hollwarth AJ, Pestell ST, Byron-Chance DH, et al. Mortality outcomes based on ASA grade in avian patients undergoing general anesthesia. *Journal of Exotic Pet Medicine* 2022;41:14-19.
84. Mans C, Sanchez-Migallon Guzman D, Lahner LL, et al. Sedation and physiologic response to manual restraint after intranasal administration of midazolam in Hispaniolan Amazon parrots (Amazona ventralis). *J Avian Med Surg* 2012;26:130-139.
85. Hawkins M, Pascoe P. Cagebirds In: West G, Heard DJ,Caulkett NA, eds. *Zoo and Wild Animal Immobilization and Anesthesia*. Ames, IA: Blackwell Publishing, 2007;269-298.
86. Edling TM. Updates on anesthesia and monitoring In: Harrison GJ,Lightfoot TL, eds. *Clinical Avian Medicine*. Palm Beach, FL: Spix Publishing, Inc., 2006;747-760.
87. Gunkel C, M. L. Current techniques in avian anesthesia. *Semin Avian Exotic Pet Med* 2005;14:263-276.
88. Muller K, Holzapfel J, Brunnberg L. Total intravenous anaesthesia by boluses or by continuous rate infusion of propofol in mute swans (Cygnus olor). *Vet Anaesth Analg* 2011;38:286-291.
89. Mulcahy DM, Tuomi P, Larsen RS. Differential mortality of male spectacled eiders (Somateria fischeri) and king eiders (Somateria spectabilis) subsequent to anesthesia with propofol,bupivacaine, and ketoprofen. *J Avian Med Surg* 2003;17:117-123.
90. Langlois I, Harvey RC, Jones MP, et al. Cardiopulmonary and anesthetic effects of isoflurane and propofol in Hispaniolan Amazon parrots (Amazona ventralis). *J Avian Med & Surg* 2003;17:4-10.
91. Hawkins MG, Wright BD, Pascoe PJ, et al. Pharmacokinetics and anesthetic and cardiopulmonary effects of propofol in red-tailed hawks (Buteo jamaicensis) and great horned owls (Bubo virginianus). *Am J Vet Res* 2003;64:677-683.
92. Machin KL, Caulkett NA. Evaluation of isoflurane and propofol anesthesia for intraabdominal transmitter placement in nesting female canvasback ducks. *J Wildl Dis* 2000;36:324-334.
93. Langan JN, Ramsay EC, Blackford JT, et al. Cardiopulmonary and sedative effects of intramuscular medetomidine-ketamine and intravenous propofol in ostriches (Struthio camelus). *J Avian Med Surg* 2000;14:2-7.
94. Machin KL, Caulkett NA. Cardiopulmonary effects of propofol and a medetomidine-midazolam-ketamine combination in mallard ducks. *Am J Vet Res* 1998;59:598-602.
95. Schumacher J, Citino SB, Hernandez K, et al. Cardiopulmonary and anesthetic effects of propofol in wild turkeys. *Am J Vet Res* 1997;58:1014-1017.
96. Lukasik VM, Gentz EJ, Erb HN, et al. Cardiopulmonary effects of propofol anesthesia in chickens (Gallus gallus domesticus). *J Avian Med & Surg* 1997;11:93-97.
97. Fitzgerald G, Cooper JE. Preliminary studies on the use of propofol in the domestic pigeon (Columba livia). *Res Vet Sci* 1990;49:334-338.
98. Kamiloglu A, Atalan G, Kamiloglu NN. Comparison of intraosseous and intramuscular drug administration for induction of anaesthesia in domestic pigeons. *Res Vet Sci* 2008;85:171-175.
99. Varner J, Clifton KR, Poulos S, et al. Lack of efficacy of injectable ketamine with xylazine or diazepam for anesthesia in chickens. *Lab Anim (NY)* 2004;33:36-39.
100. Atalan G, Uzun M, Demirkan I, et al. Effect of medetomidine-butorphanol-ketamine anaesthesia and atipamezole on heart and respiratory rate and cloacal temperature of domestic pigeons. *J Vet Med A Physiol Pathol Clin Med* 2002;49:281-285.
101. Raffe MR, Mammel M, Gordon M, et al. Cardiorespiratory effects of ketamine-xylazine in the great horned owl In: Redig PT, Cooper JE, Remple JD, et al., eds. *Raptor Biomedicine*. Minneapolis: University of Minnesota Press, 1993;150-153.
102. Ludders JW, Rode J, Mitchell GS, et al. Effects of ketamine, xylazine, and a combination of ketamine and xylazine in Pekin ducks. *Am J Vet Res* 1989;50:245-249.
103. Quandt JE, Greenacre CB. Sevoflurane anesthesia in psittacines. *J Zoo Wildlife Med* 1999;30:308-309.
104. Joyner PH, Jones MP, Ward D, et al. Induction and recovery characteristics and cardiopulmonary effects of sevoflurane and isoflurane in bald eagles. *Am J Vet Res* 2008;69:13-22.
105. Granone TD, de Francisco ON, Killos MB, et al. Comparison of three different inhalant anesthetic agents (isoflurane, sevoflurane, desflurane) in red-tailed hawks (Buteo jamaicensis). *Veterinary Anaesthesia and Analgesia* 2012;39:29-37.

106. Mercado JA, Larsen RS, Wack RF, et al. Minimum anesthetic concentration of isoflurane in captive thick-billed parrots (Rhynchopsitta pachyrhyncha). *Am J Vet Res* 2008;69:189-194.
107. Concannon KT, Dodam JR, Hellyer PW. Influence of a mu- and kappa-opioid agonist on isoflurane minimal anesthetic concentration in chickens. *Am J Vet Res* 1995;56:806-811.
108. Ludders JW, Rode J, Mitchell GS. Isoflurane anesthesia in sandhill cranes (Grus canadensis): minimal anesthetic concentration and cardiopulmonary dose-response during spontaneous and controlled breathing. *Anesth Analg* 1989;68:511-516.
109. Ludders JW, Mitchell GS, Rode J. Minimal anesthetic concentration and cardiopulmonary dose response of isoflurane in ducks. *Veterinary Surgery* 1990;19:304-307.
110. Curro T, Brunson D, Paul-Murphy J. Determination of the ED50 of isoflurane and evaluation of the analgesic properties of butorphanol in cockatoos (Cacatua spp.). *Vet Surg* 1994;23:429-433.
111. Phair KA, Larsen RS, Wack RF, et al. Determination of the minimum anesthetic concentration of sevoflurane in thick-billed parrots (Rhynchopsitta pachyrhyncha). *Am J Vet Res* 2012;73:1350-1355.
112. Naganobu K, Fujisawa Y, Ohde H, et al. Determination of the minimum anesthetic concentration and cardiovascular dose response for sevoflurane in chickens during controlled ventilation. *Vet Surg* 2000;29:102-105.
113. Zehnder AM, Hawkins MG, Pascoe PJ, et al. Evaluation of indirect blood pressure monitoring in awake and anesthetized red-tailed hawks (Buteo jamaicensis): effects of cuff size, cuff placement, and monitoring equipment. *Veterinary Anaesthesia and Analgesia* 2009;36:464-479.
114. Acierno MJ, da Cunha A, Smith J, et al. Agreement between direct and indirect blood pressure measurements obtained from anesthetized Hispaniolan Amazon parrots. *Journal of the American Veterinary Medical Association* 2008;233:1587-1590.
115. Johnston MS, Davidowski LA, Rao S, et al. Precision of repeated, Doppler-derived indirect blood pressure measurements in conscious psittacine birds. *J Avian Med Surg* 2011;25:83-90.
116. Schnellbacher RW, da Cunha AF, Beaufrere H, et al. Effects of dopamine and dobutamine on isoflurane-induced hypotension in Hispaniolan Amazon parrots (Amazona ventralis). *Am J Vet Res* 2012;73:952-958.
117. Edling TM, Degernes LA, Flammer K, et al. Capnographic monitoring of anesthetized African grey parrots receiving intermittent positive pressure ventilation. *Journal of the American Veterinary Medical Association* 2001;219:1714-1718.
118. Desmarchelier M, Rondenay Y, Fitzgerald G, et al. Monitoring of the ventilatory status of anesthetized birds of prey by using end-tidal carbon dioxide measured with a microstream capnometer. *Journal of zoo and wildlife medicine : official publication of the American Association of Zoo Veterinarians* 2007;38:1-6.
119. Schmitt PM, Gobel T, Trautvetter E. Evaluation of pulse oximetry as a monitoring method in avian anesthesia. *J Avian Med Surg* 1998;12:91-99.
120. Phalen DN, Mitchell ME, Cavazos-Martinez ML. Evaluation of three heat sources for their ability to maintain core body temperature in the anesthetized avian patient. *J Avian Med Surg* 1996;10:174-178.
121. Rembert MS, Smith JA, Hosgood G, et al. Comparison of traditional thermal support devices with the forced-air warmer system in anesthetized Hispaniolan Amazon parrots (Amazona ventralis). *J Avian Med Surg* 2001;15 187-193.
122. Boedeker NC, Carpenter JW, Mason DE. Comparison of body temperatures of pigeons (Columba livia) anesthetized by three different anesthetic delivery systems. *J Avian Med Surg* 2005;19:1-6.
123. Gunkel C, Lafortunem M. Current techniques in avian anesthesia. *Seminars in Avian and Exotic Pet Medicine* 2005;14:263-276.
124. Lichtenberger M. Shock and cardiopulmonary-cerebral resuscitation in small mammals and birds. *The veterinary clinics of North America Exotic animal practice* 2007;10:275-291.